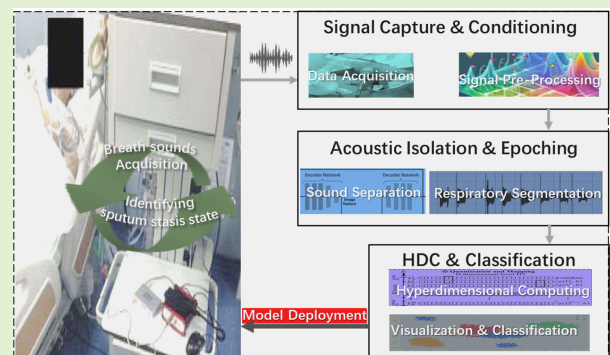


Identifying the Respiratory Sound Based on Single-Channel Separation and Hyperdimensional Computing

Jie Zheng, Yixuan Wang^{ID}, Jinglong Niu, Yan Shi^{ID}, and Fei Xie

Abstract—In intensive care units (ICUs), efficient respiratory management, particularly sputum suction in weakened patients, is critical. Traditional stethoscope-based methods for respiratory sound analysis in tracheal sputum assessment are time-consuming and often struggle to differentiate between cardiac and respiratory sounds, affecting sputum detection accuracy. To address these issues, we propose identifying respiratory sound based on single-channel separation and hyperdimensional computing (IRS-SSHC). Specifically, the proposed method first employs an encoder–decoder framework to effectively separate heart and respiratory sounds in the time domain. Then, it segments respiratory sounds using short-duration energy, where each segment is represented by a 1024-D vector space. Next, it utilizes light gradient boosting machine (LightGBM) based on the vector space for classification. Experimental results show that the classification ACC of IRS-SSHC is 97.9%, which outperforms existing methods.

Index Terms—Hyperdimensional (HD) computing, respiratory sound, single-channel separation, sound identification.



I. INTRODUCTION

IN INTENSIVE care units (ICUs), patients often require invasive ventilation, limiting their ability to voluntarily expectorate sputum [1], [2], [3]. Inability to effectively clear sputum can lead to complications such as dyspnea and lung infections [4]. Clinicians typically use pulmonary auscultation to assess sputum stasis, determining the optimal

timing for sputum expulsion. Due to the time-consuming and labor-intensive nature of this task, various researchers have developed models to identify degrees of sputum deposition. These models largely rely on clinical respiratory sound analysis, deploying training learning models into recognition systems to monitor patient sputum deposition [5], [6], [7].

At present, the prevalent approach for automatically detecting sputum accumulation is respiratory sound analysis. Naves et al. [8] utilized statistical methods to predict sputum stasis based on dominant time spectrum features, employing higher-order statistics and a divide-and-conquer strategy. Feature reduction was achieved through genetic algorithms, followed by the application of K -Nearest neighbor and naive Bayes classifiers for signal recognition [8]. However, respiratory sounds are complex, and local statistics often fail to effectively capture characteristics in both time and frequency domains. Chang and Lai proposed a method based on autoregressive and mel-frequency cepstral coefficients (MFCCs) to categorize respiratory sounds, applying dynamic time warping as a classifier [9]. The MFCC, a nonlinear cepstrum coefficient, scales linear frequency to a nonlinear scale closer to human hearing through Meir filters, reducing the sound signal in each frame within the Meir frequency scale [10].

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While high-frequency Meir filters struggle to extract effective information, and the resolution for extracting information in the energy-concentrated mid and low-frequency bands is low, posing challenges for real-time computation due to the respiratory sound signal frequency being concentrated below 2000 Hz, with acquisition rates exceeding 4000 Hz [11]. As data processing demands increase, new computing systems like hyperdimensional (HD) computing, based on HD binary vectors (HBVs), are being developed [12], [13]. These systems use high-latitude binary vectors to describe various objects, which can be an image [14] or a voice [15], [16]. The underlying computing mechanism projects this information into fixed-dimensional vector spaces, enabling effective data integration. In this context, disparate elements from sensor inputs, and other relevant data coexist within the same vector space, effectively “speaking the same language” and becoming integrable. This convergence of data into a single representational format is crucial for the next stage of processing. Leveraging this unified data representation, researchers can then utilize machine learning algorithms with greater efficacy.

An essential step in respiratory sound denoising is the separation of cardiopulmonary sounds. Heart and respiratory sound signals often overlap in frequency, necessitating time-domain methods for heart sound denoising [17], [18]. The least variance adaptive filter algorithm is commonly used, offering real-time performance without distorting breath sounds due to overlapping frequencies. Iyer et al. [17] achieved results by using electrocardiogram signals as reference signals to identify heart sound intervals. However, the difficulty in obtaining ECG signals limits the algorithm’s widespread use. Kompis and Russi [19] improved the algorithm by selecting heart sounds collected synchronously with breathing sounds as reference signals, reducing heart sound interference by 24%–49%. Overall, the above algorithms all need to introduce external signals, which are inconvenient to operate and cannot be further popularized [20]. Recent advancements in deep learning, particularly in speech and signal processing, have significantly improved multispeech recognition through techniques like mix source separation. This addresses the “cocktail-party problem,” with notable progress in deep clustering, deep attractor networks, and permutation invariant training [21], [22], [23]. Nevertheless, these techniques have not been employed in the differentiation of respiratory and cardiac sounds, owing to the absence of a dedicated training dataset.

Periodic segmentation is another critical preprocessing step [5], [24]. Traditional algorithms use short-time energy and zero-crossing rates for signal segmentation, representing time-domain signal energy and frequency-domain properties [25], [26]. These statistics, while computationally simple and less demanding in terms of respiratory sound cycle segmentation precision, are prone to noise interference, affecting signal period segmentation. To enhance accuracy, strict signal filtering is required, but this results in significant coupling between the segmentation algorithm and signal denoising. In addition, common cycle segmentation algorithms also include wavelet transform [27], [28], Viotal integral waveform method [29], time synchronous averaging [30], etc. However, the above algorithms are not applicable to the real-time moni-

toring system of sputum sound due to their complex operation or their failure to locate respiratory sound components in real time.

This article introduces a predictor, “iRSS-SCSHDC,” where mixed sounds are corrected using Chebyshev bandpass and wavelet filtering. Heart sounds are isolated in the time domain using an encoder–decoder framework, allowing for clear differentiation between heart and respiratory sounds in recording playback. Each segment is formulated through HD computing in a 1024-D vector, offering superior computational efficiency, accuracy, and robustness compared to spectrum alignment algorithms. The main contributions of this work are given as follows.

- 1) Development of a novel respiratory sound collection device: A device designed for real-time respiratory sound collection, uniquely integrating a collection and sensing structure with a patient’s ventilator through a specialized adapter, ensuring a noninvasive connection that does not disrupt the patient’s breathing or ongoing treatment.
- 2) Implementation of an encoder–decoder framework for sound separation: This work innovatively employs an encoder–decoder framework for the discrimination of respiratory sounds. The framework effectively isolates breath sounds with distinct features, crucial for accurate identification.
- 3) Utilization of HD computing in respiratory sound segmentation: The study utilizes HD computing for respiratory sound segmentation, employing the light gradient boosting machine (LightGBM) model as the classifier, and achieving an exceptionally high classification accuracy of 97.9%.

This article is organized as follows. Section II outlines the data acquisition system and methods for processing respiratory signals, including denoising, single-channel mixed sound separation, signal segmentation, HD computing, and prediction engine. Section III describes the datasets and measurement methods. Section V presents experimental results, and Section VI concludes with a summary of the iRSS-SCSHDC predictor’s effectiveness.

II. METHODS

A. Data Acquisition

In ICUs, traditional stethoscope equipment often requires direct attachment to the patient’s body surface for sound collection. This method can be challenging, especially for patients with tracheal tubes and surface wounds. To facilitate real-time, long-term monitoring without interference with other medical instruments, we have developed a novel sound sensor system with waterproof capabilities. The design involves a unique acquisition and sensing structure, seamlessly integrating with the patient’s ventilator via a specialized adapter. This innovative approach enables a noninvasive connection to the ventilator, ensuring no disruption to the patient’s breathing or ongoing treatment. As illustrated in Fig. 1(b), the device comprises six primary components: a sound sensor, an audio card, an airway connector, a waterproof connector, a signal amplifier, and a power supply unit.

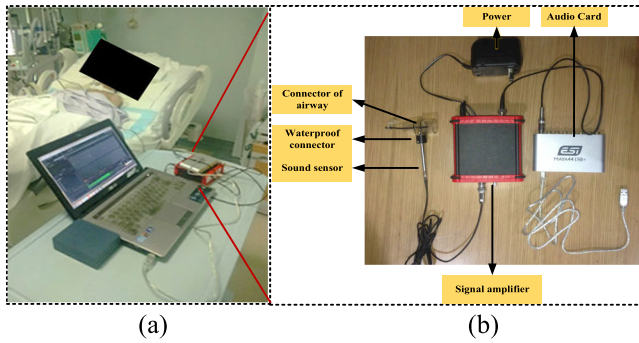


Fig. 1. (a) Respiratory sound recorded equipment. The respiratory sound signal was acquired using the captive microphone with a frequency response from 20 to 20 000 Hz and dynamic range from 30 to 126. (b) Environment of measurement.

Given the high humidity levels typical in respiratory environments due to water vapor, our system employs robust waterproofing measures to safeguard the sound sensor's functionality. Respiratory sounds are captured using a USB audio card (ESI MAYA 44, ESI Audiotechnik GmbH, Leonberg, Germany) with a sampling rate of 44 100 Hz. Fig. 1(a) showcases the device's operational setting within a clinical environment.

The weak signals emanating from the trachea are effectively amplified for clearer analysis. The device's control unit utilizes the Freescale IMX6Q chip, equipped with an onboard Linux operating system, renowned for its stable performance and suitability for concurrent computing tasks. This system records respiratory sounds onto an SD card via the secure digital input and output (SDIO) interface, storing the data in WAV file format for subsequent analysis. Based on this device, we have established the dataset2354. For a detailed description of the dataset, please refer to Section III of this article.

B. De-Noiseing and Single-Channel Mix Sound Separation

Fig. 2 demonstrates that in the complex acoustical environment of the ICU, accurate respiratory sound analysis is challenged by overlapping frequency bands of various sound sources. The phlegm sound, predominantly in the range of 20–20 000 Hz, overlaps with the heart sound in the low-frequency band (20–600 Hz) [31], [32]. Our approach employs a Chebyshev bandpass filter with a passband of 20–2000 Hz, effectively filtering out noise components that do not overlap with the sputum sound frequency band, and data explosion in pulmonary acoustic high-frequency band is solved. An enhanced wavelet-filtering algorithm is then applied to eliminate main noise components, including environmental noise and human voice, while preserving the integrity of the sputum sound [33]. Despite these initial steps, the critical challenge remains in distinguishing between heart and respiratory sounds, as their frequency bands significantly overlap in the low-frequency range.

To address this problem, we adopt a novel mix source separation approach, predicated on several assumptions [34], [35], [36]. We consider C as the number of sources in the mixed sound $x(t)$, segmenting the sound into K distinct parts,

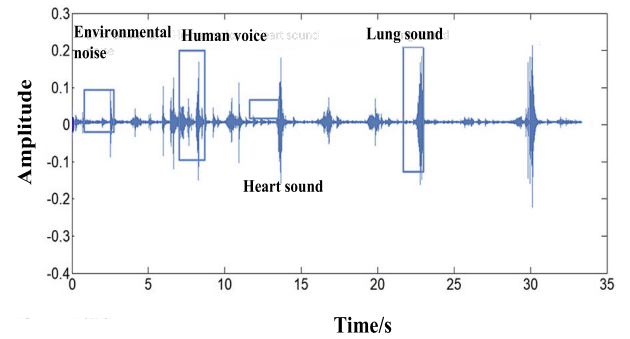


Fig. 2. Clip of respiratory sound.

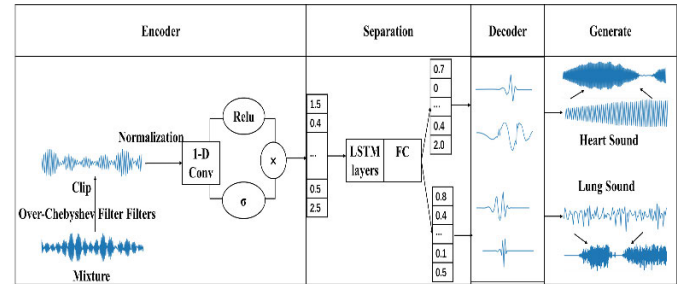


Fig. 3. Structure of the network. It consists of four parts: encoder module, separation module and decoder module, and generation module. The encoder module and the separation module are used to generate the mixed weight matrix and the feature matrix corresponding to each source, and the decoder module is used to decode the 1-D convolution to generate the base signal. The generating module is used to generate two source signals of heart sound and respiratory sound.

and then assume that the sound has N basis signals, which are marked as $B = [b_1, b_2, \dots, b_N]$. Assume that each source can be reformulated by multiplying the base signal B with their weight matrix w_i , $i \in [1, 2, \dots, C]$. Then the mix source separation can be converted into finding the set basis signal B and the weight matrix w_i corresponding to each signal source.

Implementing a deep learning strategy, we aim to separate respiratory and heart sounds, as demonstrated in the network structure shown in Fig. 3. The network comprises three critical modules: encoder, separation, and decoder. Mixed sounds are inputted to the encoder module where features are extracted via 1-D convolution. These features are multiplied by two activation functions (Relu and δ activation functions) to synthesize the weight matrix $W (W = w_1 + w_2)$ of the mixture sound over basis signals B [37]. LSTM layers and a fully connected function restore the feature matrices of the two source signals, while the decoder module retrieves the basis signals through deconvolution over the 1-D convolution kernel of the encoder module [38]. The resultant feature matrix is combined with the basis signal to derive the predicted heart and respiratory sound signals. The network optimization employs SI-SNR as the loss function and Adam as the optimizer [39].

C. Respiratory Segmentation

Each signal corresponds to one breath cycle, which includes both inspiration and expiration phases, as illustrated in Fig. 4. In this study, we have segmented each data sample into two distinct phases: inspiration and expiration. As observed

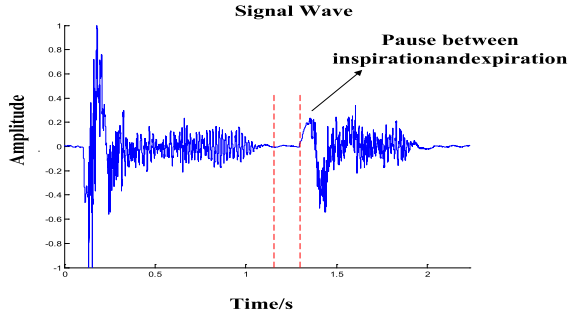


Fig. 4. One respiratory cycle last 2.5 s. The inspiratory time lasts about 1 s and expiratory time lasts about 0.7 s. The pause time lasts about 0.15 s. The pause between inspiration and expiration is much shorter than the time of inspiration and expiration.

from Fig. 4, a clear pause exists between the inspiration and expiration phases, with energy predominantly concentrated during these two phases

Based on the short-time average energy, the respiratory signal is segmented into inspiratory and expiratory phases. The short-time energy represents the energy within a single segment or across multiple segments of the respiratory cycle. This concept is defined as follows:

$$E_i = \sum_{m=1}^M x_i(m)^2 \quad (1)$$

where $x_i(m)$ is the i th frame discrete sound signal, and M is the length of frame. Two thresholds E_{th}^1 and E_{th}^2 are set— $E_{th}^1 = aE_c$, $E_{th}^2 = bE_c$ ($E_{th}^1 > E_{th}^2$). When $E_i > E_{th}^1$, the segment is classified as part of the inspiration or expiration phase. Furthermore, if the condition where $E_i < E_{th}^2$ persists for a duration shorter than the maximum silence period (defined in this study as 250 frames), the corresponding frame is also considered as belonging to the inspiration or expiration phase. However, the variable nature of the measurement environment and patient-specific factors make the energy of respiratory sound signals inherently dynamic. This variability presents a challenge in selecting a constant threshold for reliably determining the categorization of each frame as either inspiration or expiration.

In this study, we employ an iterative algorithm to automatically determine the threshold. The algorithm initializes with E_c , the mean energy of noise frames devoid of any breathing sounds. The parameters a and b are set to 2 and 1, respectively. Considering the breathing characteristics of patients, which align with the ventilatory assistance provided, we assume that the pause time between inspiration and expiration is significantly shorter than the duration of either phase. Furthermore, we posit that each sample data comprises just two start points and two endpoints, facilitating effective segmentation. The procedure for this segmentation process is detailed in Fig. 5.

Fig. 6 demonstrates the outcomes of the segmentation process. It is evident that the inspiration and expiration phases are accurately distinguished. In this study, we specifically focus on the initial 0.6 s of both the inspiration and expiration phases for feature extraction. This approach is adopted due to the concentration of key sputum-related features predominantly occurring at the beginning of these respiratory phases.

D. HD Computing of Respiratory Sound

In this study, we introduce a novel method for mapping these respiratory sound segments into a vector space. This process utilizes HD computing, facilitating adaptive learning of sound segments and enabling real-time inference for segment classification [13]. HD computing, an emerging paradigm, is known for its robustness to noise and errors and its capacity for rapid and straightforward learning [12], [40]. It operates by projecting information or tasks into a high-dimensional vector space (referred to as a supervector space) and then employing simple computational operations for classification or inference [12], [40]. We apply this method specifically to respiratory sound clip training, integrating it with an HD computing algorithm for effective inference. The features of the preprocessed sound clips, characterized by their continuous values and low dimensionality, are projected into a randomly initialized binary vector of 1024 dimensions. This projection involves combining the input signal with a HD vector from the vector table. The corresponding binary vector for the input signal is identified using the Hamming distance. By aggregating the corresponding elements of all vectors, we obtain a 1024-D representation of the sound clip.

E. Prediction Engine

Upon representation of respiratory sound clips as fixed-dimensional vectors, the utilization of machine learning models for outcome prediction becomes highly effective. Fig. 6 showcases the area under curve (AUC) results for various machine learning methodologies, including LightGBM, gradient boosted decision tree (GBDT), random forest (RF), and deep neural network (DNN). In these tests, the LightGBM method exhibited a particularly high AUC, leading to its selection as the primary prediction model. Fig. 7 outlines the flowchart of our iRSS-SCSHDC model, which processes ground and respiratory sounds through a sequence of critical transformations: denoising, separating heart and respiratory sounds, segmentation, and encoding into high-dimensional vectors. Denoising ensures signal purity, while the separation step isolates overlapping frequencies, preventing heart sound interference that could lead to segmentation errors. Segmentation breaks down the audio into manageable cycles, and encoding captures the essential sound characteristics. Finally, the LightGBM predictor uses these features to generate precise classifications, significantly improving diagnostic accuracy and robustness in clinical settings.

III. DATASETS

With the respiratory sounds measuring system, we collected two datasets. The newly collected dataset is called the “dataset2354,” in which 2354 groups of sputum sound data were involved, and each group of data contained one sputum sound signal, one pressure signal, and one flow signal. All the signals were collected from 1177 patients, and two sets of data were collected for each patient. Data were collected from the same patient with a time interval of more than 8 h, and the collection was done with the consent of the patient or, if the patient was a minor, with the consent of their

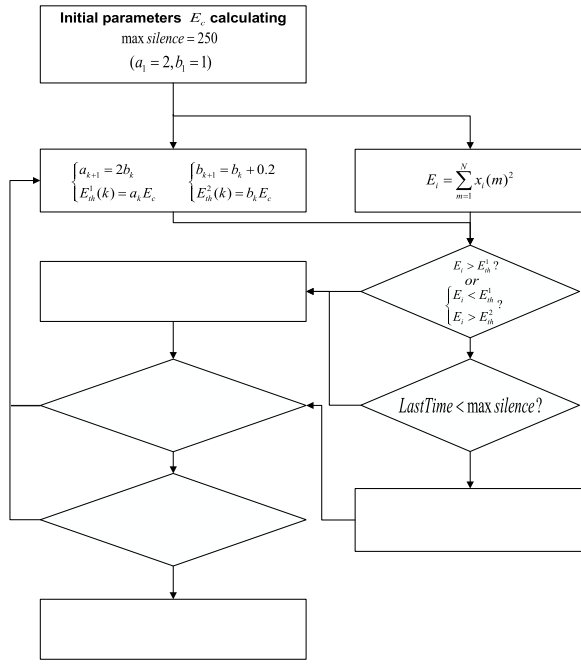


Fig. 5. Procedure of segmentation. The method based on the characteristics of breathing cycle search the appropriate the threshold E_{th}^1 and E_{th}^2 .

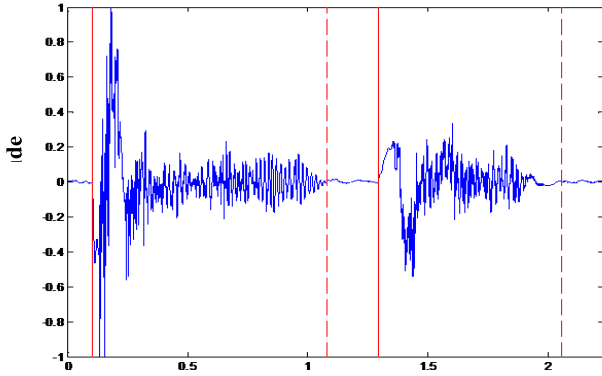


Fig. 6. Segmentation. The solid red lines mark the start of the inspiration or expiration and dashed red dashed lines mark the end of the inspiration or expiration.

guardian. During data collection, the labeling of sputum sound data depends on the judgment of medical staff. The clinician determines the degree of sputum stasis by stethoscope, and then carries out sound collection and marks the degree of sputum stasis. All annotated data were judged by clinicians twice, and distorted data and missing data of physiological characteristics of sputum sound were eliminated, so as to ensure that the data involved in training could judge the degree of sputum deposition of patients. The training benchmark dataset S is defined by

$$S = S^+ + S^- \quad (2)$$

where S^+ is the set of sputum sounds and S^- is the set of nonsputum sounds, the sputum sound data information is shown in Table I.

TABLE I
SPUTUM SOUND DATA INFORMATION

Sputum Stasis Degree	Number of Patients	Number of Original Signals	Number of Signals After Segmentation	Age, years	Sex, % male
0	223	446	5317	67.4 ±8.6	57.9
1	287	574	6882	72.6 ±12.3	52.6
2	288	576	7771	69.5 ±15.1	45.7
3	379	758	9898	75.4 ±10.4	53.9
Total	1177	2354	29868	--	--

The second dataset is called the benchmark dataset, in which 840 respiratory sound samples were collected from 14 patients, with each sample corresponding to one respiratory cycle, taken from [7].

Since the data for training the single-channel mix source separation network is called cardiopulmonary murmur dataset, which is collected from the network, it contains 200 heartbeat sounds and 380 breath sounds collected from electronic stethoscope equipment. By analyzing the data from a benchmark dataset, the mixture ratio of heart sounds and respiratory sounds was determined to be 0.3, based on calculating their average energy ratio, finally, 7600 cardiopulmonary murmur sounds were generated. Using mixed proportions to generate datasets can simulate this clinical environment, thereby improving the validity and robustness of the model in real-world applications.

IV. MEASUREMENT

To determine a segment as a sputum sound or nonsputum sound belongs to the case of single-label classification, the proposed predictor can be evaluated by the metrics such as accuracy (Acc), sensitivity (Sn), and specificity (Sp), the formulas are as listed below

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$Sn = \frac{TP}{TP + FN} \quad (4)$$

$$Sp = \frac{TN}{TN + FP} \quad (5)$$

where TP is the number of the sputum sound with correct prediction, TN is the number of the nonsputum sound with correct prediction, FP is the number of the nonsputum sound pair with misprediction, and FN is the number of the sputum sound with misprediction.

V. RESULTS AND DISCUSSION

We compared the influence of the mix source separation network on the dataset2354. The noise components that do not overlap with the frequency band of the sputum sound are filtered through the Chebyshev filter; the separation of the

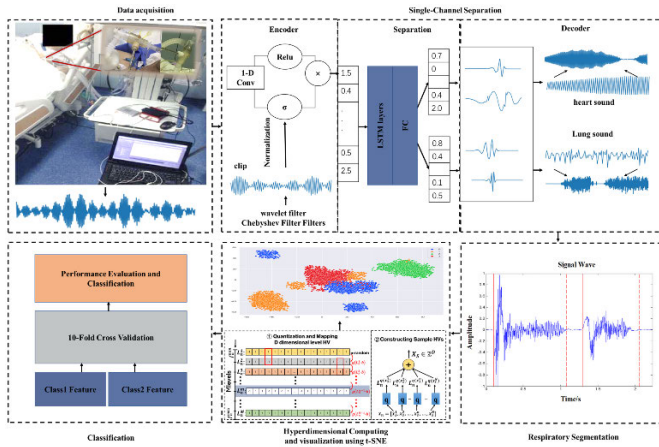


Fig. 7. Flowchart of the iRSS-SCSHDC model.

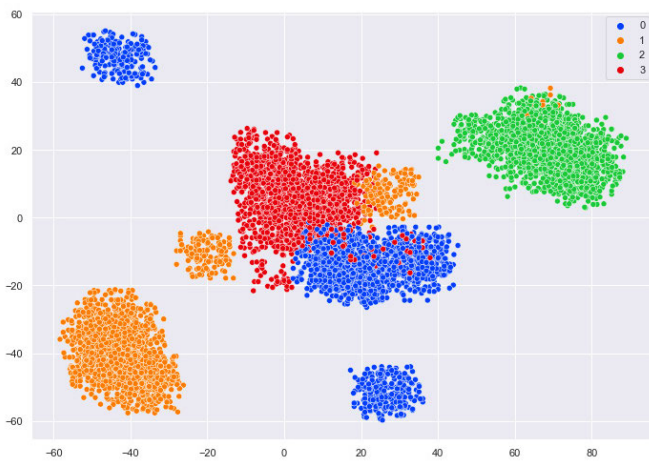


Fig. 8. Hypervectors in 2-D for identifying the respiratory sound signal using t-SNE. Each point represents a sample, with 0 represent expiration sputum sound and 1 represent expiration nonsputum sound, 2 represent noninspiration sputum sound and three inspiration represent sputum sound.

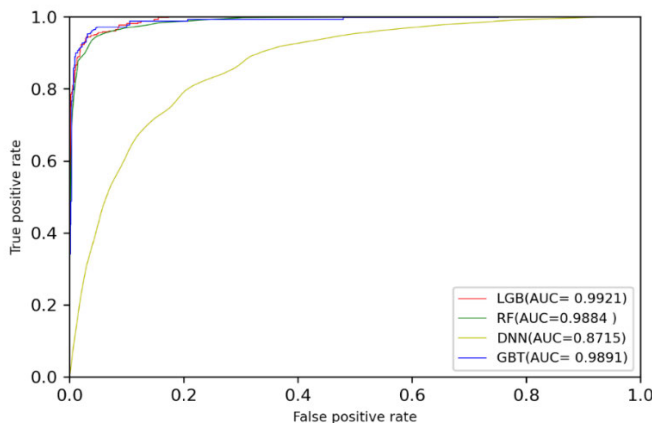


Fig. 9. ROC curves with different models the dataset2354.

heart sound components is realized through the mix source separation algorithm of a single-signal source, as Table II shows, the experimental results obtained by heart-respiratory sound separation are better, the accuracy, sensitivity, specificity values have been significantly improved. After the clinician's

TABLE II
INFLUENCE OF THE MIX SOURCE SEPARATION NETWORK
ON THE DATASET 2354

Mix sound separation	Inspiration			Expiration		
	Sen(%)	Prec(%)	Acc(%)	Sen(%)	Prec(%)	Acc(%)
Without	91.65	90.18	90.09	89.90	89.11	89.50
With	98.69	98.01	98.45	97.68	97.02	97.35

TABLE III
DIFFERENT HD VECTOR DIMENSIONS RESULTS OF THE
PROPOSED MODEL

HD dimension	Inspiration			Expiration		
	Sen(%)	Prec(%)	Acc(%)	Sen(%)	Prec(%)	Acc(%)
512D	96.95	95.28	96.10	95.96	95.05	95.50
1024D	98.79	98.11	98.45	97.68	97.02	97.35
2048D	98.92	98.32	98.62	99.29	97.93	98.60

auscultation judgment, the separated sputum sound signal has not lost its acoustic characteristics, and the medical staff can still judge the degree of sputum accumulation in the patient's body from the filtered signal.

To achieve an efficient balance between calculation dimension and reduced runtime in HD vector encoding, we experimented with representing sound clips in dimensions of 512-D, 1024-D, and 2048-D. As indicated in Table III, the accuracy (ACC) value for the 512-D vector was 2.35% lower than that of the 1024-D vector. Notably, both the 1024-D and 2048-D vectors significantly enhanced classifier performance, with similar ACC values. However, to minimize computational demands, we opted for the 1024-D dimension to represent respiratory sounds, the total accuracy of inspiration and expiration was 97.9%. Fig. 8 presents the hypervectors in a 2-D format, utilizing the t-distributed stochastic neighbor embedding (t-SNE) algorithm. This visualization clearly illustrates that different sound classes can be effectively distinguished and separated.

To validate the effectiveness of our method in separating heart and respiratory sounds, we conducted an auditory analysis by playing back the recorded sounds. This allowed us to distinctly hear the separation between heart and respiratory sounds. Subsequently, we extracted pure breathing sounds and phlegm sounds from these recordings. The clarity of these extracted sounds provides a valuable tool for medical staff to assess and make informed judgments regarding sputum discharge in patients. Fig. 9 displays the comparative performance of various algorithms, with the default classifier LightGBM exhibiting superior generalization ability compared to RF, DNN, and GBDT. In comparison with methodologies discussed in previously published articles, our feature extraction method demonstrates remarkable accuracy, sensitivity, and specificity [5].

Table IV shows the performance of iRSS-SCSHDC in the benchmark dataset [7], we can find that separation of

TABLE IV
COMPARED WITH PREVIOUS WORK ON BENCHMARK DATASET

Feature set	Inspiration		Expiration	
	Sensitivity	Specificity	Sensitivity	Specificity
EE+ MSV[7]	87.10	85.20	75.70	84.00
EE+F[7]	91.60	92.50	88.60	85.20
MSV+F[7]	90.30	91.00	85.90	84.00
EE+ MSV+F[7]	92.10	91.70	86.10	85.20
HDC+F	85.55	88.36	87.09	85.93
HDC+LGB	89.90	89.11	87.40	86.88
SCS+HDC+F	96.45	95.26	96.27	95.54
SCS+HDC+LGB	97.88	96.93	97.17	96.14

heart sounds is important to improve the performance of sputum detection. Moreover, sensitivity and specificity values of inspiration not very far from expiration, indicating that iRSS-SCSHDC has higher accuracy and robustness.

VI. CONCLUSION

In this study, we introduced the identifying respiratory sound based on single-channel separation and HD computing framework, a novel approach that addresses critical challenges in respiratory management within ICUs. By effectively separating heart and respiratory sounds using an encoder-decoder framework, and leveraging HD computing for sound segmentation, the IRS-SSHC framework demonstrated a significant improvement in classification ACC. The development of a specialized respiratory sound collection device, integrated seamlessly with a patient's ventilator, represents a key innovation, providing a noninvasive and reliable solution for real-time sound collection. This work not only advances the current state of respiratory sound analysis but also lays the groundwork for future innovations in this field. Future research could focus on developing quantifiable metrics for assessing sputum retention levels, which would provide a more precise tool for clinical use. Moreover, extending this research into the intelligent diagnosis of respiratory diseases such as pneumonia and bronchitis, through the recognition of sputum sounds, could greatly enhance its clinical relevance.

In conclusion, the IRS-SSHC framework offers a step forward in the accurate and efficient detection of respiratory sounds, providing a powerful tool for clinical settings that could ultimately improve patient outcomes in critical care environments.

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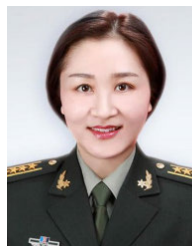
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